

Task 5 : Logic Building task

Introduction

Logic Building project focuses on designing and implementing a logic-based machine learning system that translates English words into Hindi with specific operational constraints. The application emphasizes logic building, rule-based validation, and controlled access based on time and linguistic conditions. The goal of this project is not only translation accuracy but also the application of real-world constraints to machine learning systems.

The project integrates machine learning techniques with a graphical user interface to create an interactive and user-friendly translation tool. By combining ML, Python programming, and GUI development, the project demonstrates how intelligent systems can be built with conditional logic and user guidance mechanisms.

Background

Language translation systems are widely used in modern applications, but most systems do not incorporate logical restrictions based on linguistic patterns or time-based rules. This project was developed to strengthen logical reasoning skills by introducing vowel-based restrictions and time-window access for translations.

The system is trained using a custom English–Hindi dataset and applies character-level TF-IDF vectorization along with a Logistic Regression model. The integration of fallback similarity logic ensures reasonable outputs even when model confidence is low, enhancing robustness.

Learning Objectives

The primary learning objective of this project is to understand how machine learning models can be combined with rule-based logic to enforce constraints beyond pure prediction. The project helps in learning how to validate inputs, manage model confidence, and handle exceptions gracefully.

Another objective is to gain hands-on experience in developing end-to-end ML applications, from dataset preparation and model training to GUI deployment. This also includes learning file handling using pickle and integrating trained models into real-time applications.

Activities and Tasks

The project involved collecting and preprocessing an English–Hindi word dataset, removing null values, and transforming text into numerical features using TF-IDF vectorization. A Logistic Regression model was then trained to predict Hindi translations based on English input words.

Additional tasks included implementing vowel-based validation rules, time-based access logic, and confidence-based fallback mechanisms using cosine similarity. A Tkinter-based GUI was developed to allow users to input words and view translations interactively.

Data :

	English	Hindi
0	bell	घंटी
1	roadmaster	रोडमास्टर
2	overinterest	अतिरुचि
3	hemokonia	hemokonia
4	synclitism	समकालिकता
...	...	...
495	phonological	ध्वनी
496	grammatist	व्याकरणशास्त्री
497	gasconader	गैसकोनेडर
498	writative	लेखनात्मक
499	Berninesque	बर्निनेस्क

500 rows × 2 columns

Model :

```
# Features (X) and labels (y)
X = df['English'].astype(str)
y = df['Hindi'].astype(str)

# Convert English text into numerical features
vectorizer = TfidfVectorizer(analyzer='char', ngram_range=(1, 3))
X_vec = vectorizer.fit_transform(X)

Train the Logistic Regression model

# Train ML model
model = LogisticRegression(max_iter=1000)
model.fit(X_vec, y)
```

LogisticRegression
LogisticRegression(max_iter=1000)

Testing :

```
while True:
    word = input("Enter English word (or 'exit' to quit): ").strip()
    if word.lower() == 'exit':
        break
    print(translate_word(word))

Enter English word (or 'exit' to quit): bell
Translation: bell → घंटी
Enter English word (or 'exit' to quit): roadmaster
Translation: roadmaster → रोडमास्टर
Enter English word (or 'exit' to quit): phonological
Translation: phonological → ध्वनी
Enter English word (or 'exit' to quit): exit
```

Skills and Competencies

This project enhanced skills in Python programming, particularly in machine learning libraries such as scikit-learn, NumPy, and Pandas. It also strengthened understanding of text vectorization techniques and classification models.

In addition, the project improved competencies in GUI development using Tkinter, logical problem-solving, and integrating ML models into user-facing applications.

Feedback and Evidence

The system provides immediate feedback to users through GUI-based error messages when invalid inputs are entered or when vowel-based time restrictions are violated. This ensures clarity and improves user interaction with the application.

Evidence of successful implementation includes correct translations during valid conditions, proper error handling for restricted cases, and smooth GUI operation. The trained model file and consistent outputs serve as proof of effective system design.

GUI :



The screenshot shows a web application window titled "English → Hindi Translator (ML Model)". The main heading is "English → Hindi Translator". Below it, there is a label "Enter English Word:" followed by a text input field. A green button labeled "Translate" is positioned below the input field. Underneath the button is the label "Hindi Translation:" followed by another text input field. At the bottom of the window, a note states: "Note: Words starting with vowels can only be translated between 9 PM and 10 PM."

GUI Working :

English → Hindi Translator (ML Model)

English → Hindi Translator

Enter English Word:

Translate

Hindi Translation:

Note: Words starting with vowels can only be translated between 9 PM and 10 PM.

English → Hindi Translator (ML Model)

English → Hindi Translator

Enter English Word:

Translate

Hindi Translation:

Note: Words starting with vowels can only be translated between 9 PM and 10 PM.

GUI Error :



Challenges and Solutions

One of the main challenges faced was handling low-confidence predictions from the machine learning model. This was resolved by implementing a cosine similarity fallback mechanism that selects the closest matching word from the dataset.

Another challenge was enforcing time-based and vowel-based logic without disrupting the user experience. This was addressed by integrating clear validation checks and informative GUI alerts, ensuring that users understand why certain translations are restricted.

Outcomes and Impact

The project successfully demonstrates a logic-driven machine learning application that goes beyond standard translation functionality. It effectively combines ML predictions with real-world constraints, improving logical reasoning and system control.

The outcome is a reliable English-to-Hindi translator that enforces vowel and time-based rules while maintaining usability. This project serves as a strong foundation for more advanced language processing and rule-based AI systems.

Conclusion

In conclusion, this logic-building translation project highlights the importance of integrating machine learning with rule-based validation and user-centric design. The project meets all specified requirements, including custom ML training, GUI implementation, and controlled translation behavior.

Overall, the project provides valuable hands-on experience in developing intelligent systems with constraints, preparing the foundation for more complex AI applications. It successfully achieves its learning and functional objectives while maintaining clarity and robustness.